

Solution Requirements

Date	28 June 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low) 0
1	Authentication	Users securely log in using username/email and password. Admin authentication is required to manage the system.	High
2	Authorization levels	Admin can manage users, datasets, and prediction history. Users can submit applicant details and view prediction results only.	High

3	External interfaces	Integrates with Flask web application, Machine Learning model (Pickle), IBM Cloud deployment, and CSV dataset storage	High
4	Transactions processing	User enters applicant details, system validates input, processes data through the ML model, and returns Approved or Rejected instantly.	High
5	Reporting	Displays prediction results, stores prediction history, and generates reports showing approval/rejection statistics.	Medium
6	Business rules, etc.	Credit card approval is based on applicant income, employment status, credit history, age, and other financial attributes. Invalid or incomplete inputs are rejected.	High
7	Compliance to laws or regulations	User data must be securely stored and handled according to data privacy standards. Sensitive information should not be exposed.	High
8	<i>Other</i>	System provides user-friendly interface, input validation, error handling, and clear prediction explanations.	Medium

Step 2: Non-Functional Requirements (NFR)

1	Performance & Speed	The system should process predictions quickly.	Prediction response time < 2 seconds
2	Scalability	System should support multiple users simultaneously without performance degradation.	Supports up to 500 concurrent users
3	Security & Data Privacy	Encrypt user data, secure login, and protect against unauthorized access.	HTTPS enabled, encrypted passwords, secure data storage
4	Reliability & Availability	System should remain available with minimal downtime	99.9% uptime
5	Usability & Accessibility	Easy-to-use interface that works on desktop and mobile browsers.	Responsive design and intuitive UI

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

6	<i>Other</i>	Maintainable, portable, and easy to update with new ML models or datasets.	Modular code with easy deployment and maintenance
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